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Adaptive Multi-scale Aggregation with Residual Attention for Remote Sensing Salient Object Detection

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***Abstract*—Salient object detection (SOD) in remote sensing imagery plays a critical role in complex scene understanding ,but remains highly challenging due to large-scale variability, weak target-background contrast, and noisy contextual information. To address these issues, we propose AMARA-Net (Adaptive Multi-scale Aggregation with Residual Attention Network), which integrates residual channel attention with adaptive multi-scale global context aggregation. The residual channel attention mechanism enhances feature discrimination by emphasizing informative responses and suppressing noise. Meanwhile, the adaptive multi-scale aggregation module employs dilated convolutions with varying receptive fields to capture fine local details and model global semantic dependencies, improving robustness to scale variation and background clutter. An attention-guided fusion strategy further harmonizes residual and multi-scale representations for complementary feature learning. Extensive experiments on multiple remote sensing benchmarks demonstrate that AMARA-Net achieves state-of-the-art performance, delivering higher accuracy, better structural integrity, and sharper boundary delineation. These results validate the effectiveness of jointly modeling residual attention and adaptive multi-scale context aggregation for robust salient object detection in remote sensing images.**

***Index Terms*—Residual Channel Attention,Salient Object Detection (SOD),Adaptive Multi-Scale Aggregation,Remote Sensing Image (RSI).**

I. Introduction

Salient object detection (SOD) in remote sensing imagery is fundamental for applications such as land cover classification, urban planning [1], environmental monitoring, and disaster assessment. By highlighting distinctive targets from complex large-scale scenes, SOD supports advanced vision tasks including change detection and scene understanding.

Despite the progress achieved through deep learning, existing methods still face two main challenges: (1) reliance on single-scale or shallow aggregation strategies that limit the ability to capture objects of varying sizes; and (2) the coupling of salient and background features, which introduces noise, blurred boundaries, and incomplete detection in cluttered environments.

To address these issues, we propose AMARA-Net (Adaptive Multi-scale Aggregation with Residual Attention Network), which integrates Residual Channel Attention Blocks (RCAB) with an Adaptive Multi-scale Global Context Aggregation (AMGCA) module. RCAB enhances feature discrimination by emphasizing informative channels and suppressing noise, while AMGCA employs dilated convolutions with multiple receptive fields to jointly capture fine-grained details and global context. An attention-guided fusion strategy further harmonizes these features for accurate localization and boundary refinement.

The main contributions of this work include an adaptive multi-scale aggregation framework that effectively models both local and global dependencies, a residual attention mechanism that strengthens salient feature representation, and an attention-guided fusion strategy that ensures robust and boundary-aware detection in complex remote sensing scenes.

II. Related Work

A. Remote Sensing Image Salient Object Detection

Salient object detection (SOD) in remote sensing imagery has been extensively studied due to its importance in land use analysis [2], environmental monitoring, and urban planning.The emergence and continuous evolution of deep learning

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have significantly driven the development of convolutional neural networks (CNNs) in visual understanding tasks. Architectures based on an encoder–decoder paradigm, such as U-Net and fully convolutional networks, introduced multi-scale fusion mechanisms, while later works incorporated dilated convolutions and pyramid pooling modules to enlarge receptive fields and jointly capture local details and global semantics.

Attention mechanisms, including channel and spatial attention, further improved discriminative feature learning by emphasizing informative responses and suppressing background noise. Residual learning strategies also contributed to network stability and deeper feature extraction. However, these CNN-based approaches still face difficulties in balancing residual enhancement, multi-scale feature integration, and noise suppression, which remain open challenges in remote sensing SOD.Recently, transformer-based architectures have shown promise by modeling long-range dependencies and global relationships. Methods such as TransSOD and ViT-based SOD frameworks improve contextual reasoning [3] but often suffer from high computational cost and insufficient local detail preservation. In parallel, weakly supervised SOD approaches have emerged to reduce annotation costs but typically compromise boundary precision and robustness, especially in complex remote sensing scenes. These limitations highlight the need for models that can efficiently integrate both local and global cues while maintaining noise resilience.

B. Attention Mechanisms and Multi-scale Context Modeling in Remote Sensing SOD

Attention mechanisms enhance feature discrimination by emphasizing salient responses and suppressing irrelevant background information. The Squeeze-and-Excitation (SE) block and its derivatives improve channel weighting, while spatial attention focuses on key regions. The Residual Channel Attention Block (RCAB), initially developed for image restoration, effectively refines channel responses while preserving residual learning for stable optimization. In remote sensing SOD, such mechanisms strengthen object saliency but are often insufficient in handling large-scale variation and background clutter.

Multi-scale modeling is thus essential for robust SOD. Approaches such as pyramid pooling, ASPP, and dilated convolutions have been proposed to aggregate features across diverse receptive fields. While these designs expand contextual understanding, naïve fusion can introduce redundancy and noise, leading to degraded detection quality. Transformer-based multi-scale designs partially alleviate this issue through hierarchical token interaction, yet they still face challenges in balancing global context and local precision.

To overcome these shortcomings, we propose AMARA-Net (Adaptive Multi-scale Aggregation with Residual Attention Network), which integrates RCAB with an Adaptive Multi-scale Global Context Aggregation (AMGCA) module. RCAB enhances feature discrimination by emphasizing informative channels and suppressing noise, while AMGCA employs dilated convolutions with adaptive receptive fields to jointly capture fine-grained and global contextual cues. An attention-guided fusion mechanism further harmonizes residual and multi-scale features, enabling robust and noise-resistant salient object detection in complex remote sensing imagery.

III. METHOD

A. Overall Network Architecture

The designed model adopts an encoder–decoder framework, as shown in Figure.1. The encoder adopts the Pyramid Vision Transformer v2 (PVTv2-B2) as backbone to extract hierarchical multi-scale features F1,F2,F3,F4 with increasing semantic abstraction and decreasing spatial resolution. To enhance feature representation, each stage is equipped with Residual Channel Attention Blocks (RCAB) are employed to adaptively highlight informative feature channels, while maintaining residual pathways to ensure stable and efficient optimization.

In order to tackle the issues of multi-scale objects and background clutter in remote sensing images, we design a Parallel RCAB–AMGCA Attention Module (PRAM), which jointly processes features through RCAB and Adaptive Multi-scale Global Context Aggregation (AMGCA). Specifically, AMGCA aggregates global context features from multiple receptive fields in an adaptive manner, thereby capturing both long-range dependencies and local discriminative structures. By introducing adaptive weighting, AMGCA suppresses redundant background information and emphasizes salient object cues across different scales. The outputs from RCAB and AMGCA are then fused via an attention-guided mechanism, ensuring complementary integration of residual refinement and multi-scale global context modeling.

On the decoder side, we progressively upsample semantic-rich features and combine them with detail-preserving low-level details via skip connections. Each fusion stage is further refined by PRAM to suppress noise and enhance discriminative saliency cues. Finally, a prediction head generates the saliency map, which is upsampled to the original resolution. This hierarchical integration of residual attention and adaptive global context aggregation enables our network to achieve robust salient object detection with precise localization and sharp boundary preservation in remote sensing imagery.

B. Overall Network Architecture

To refine the descriptive quality of salient object features within remote sensing data, we adopt the Residual Channel Attention Block (RCAB) as a fundamental building unit in our network. The RCAB is designed to exploit inter-channel dependencies and strengthen informative features while suppressing less useful ones.

Formally, given an input feature map $X \in R^{B\times C\times H\times W}$, the RCAB performs two successive convolutional operations followed by a Channel Attention (CA) mechanism. Each convolution uses a kernel size of 3 and preserves the spatial resolution via appropriate padding. Batch normalization and

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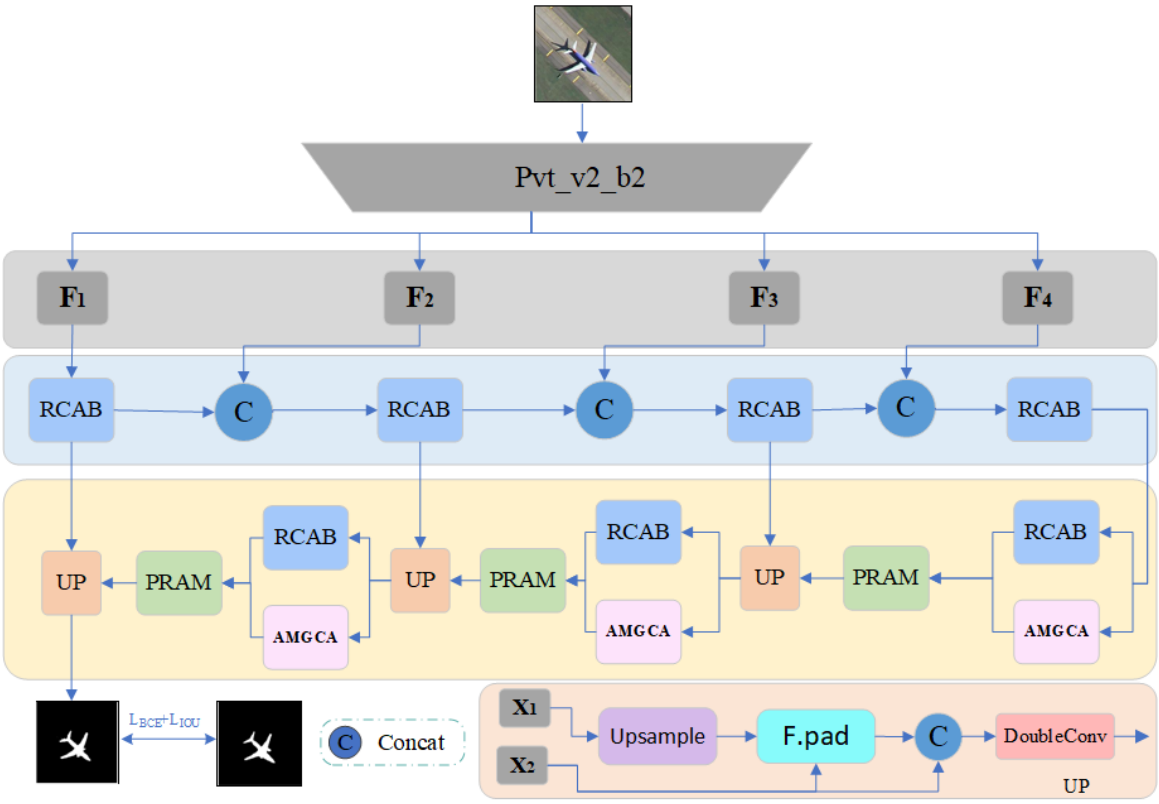


Figure 1: Overall framework of AMARA-Net.

ReLU activation are optionally applied after the first convolution to stabilize training and introduce non-linearity.

The channel attention mechanism is realized through the use of global average pooling, subsequent to a two-stage convolutional network, which first reduces the channel dimension by a factor of r and then restores it to the original dimension. A sigmoid activation generates a channel-wise attention map Mc, which is applied to the input features in a residual fashion:

$$Y = X + M_c(F(X)) \tag{1}$$

where F(X) represents the sequential convolutional operations. The residual connection ensures gradient propagation and stabilizes training, while the attention mechanism adaptively emphasizes salient features relevant to the target objects in high-resolution remote sensing imagery.

In our proposed network, RCABs are inserted after multi-level feature aggregation to refine intermediate representations before further multi-scale attention processing, enabling precise localization of object boundaries and improving overall detection performance.

C. Overall Network Architecture

In remote sensing images, salient objects commonly appear with substantial changes in scale and are surrounded by noisy or cluttered backgrounds, which makes single-scale feature modeling insufficient. To address this, we design the Adaptive Multi-scale Global Context Aggregation (AMGCA) module to adaptively model both detailed local structures and global semantic relationship.

Given an input feature map $X \in R^{C\times H\times W}$, AMGCA first employs multiple parallel dilated convolutionsd1,d2,. . . ,dk with varying dilation factors to aggregate contextual cues across different scales:

$$F_i = Conv_{d_i}(X) \quad i = 1, 2, ..., k \tag{2}$$

where $Conv_{d_i}()$ denotes a convolution with dilation rate d_i. These branches enlarge the receptive field without significantly resulting in increased computational burden, enabling the module to perceive both local and global context.

Subsequently, the multi-scale features are concatenated and projected into a unified representation:

$$F_{ms} = \phi(Concat(F_1, F_2, ..., F_k)) \tag{3}$$

where ϕ is a 1×1 convolution followed by batch normalization and ReLU activation.

To adaptively suppress redundant or noisy responses, AMGCA employs a global context-aware attention mechanism. Specifically, a channel-wise attention weight vector $\alpha \in R^c$ is generated by global average pooling followed by a gating function:

$$\alpha = \sigma(W_2\delta(W_1 GAP(F_{ms}))) \tag{4}$$

where GAP() denotes global average pooling, W1 and W2 are learnable weights, σ is the ReLU activation, and δ is the

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sigmoid function. Finally, the adaptive aggregated feature is obtained as:

$$F_{amgca} = \alpha \odot F_{ms} \tag{5}$$

where $\odot$ denotes channel-wise multiplication. This process selectively emphasizes salient features across scales while suppressing background noise, yielding a robust multi-scale representation.

D. Loss Function

To refine both local and global learning objectives, training is guided by a composite supervision strategy that integrates Binary Cross-Entropy and IoU-based losses. BCE supervises pixel-wise classification and enforces structural consistency of salient regions, while IoU explicitly measures region overlap to refine localization accuracy. Their complementary roles yield a balanced objective that enhances both global integrity and boundary precision of salient objects. The overall loss is defined as:

$$loss = bce_loss(sal, gts) + iou_loss(\sigma(sal), gts) \tag{6}$$

IV. Experimental Results

A. Datasets

For a rigorous evaluation, we conducted experiments on ORSSD (800 images: 600 training, 200 testing) and its extended counterpart EORSSD (2,000 images: 1,400 training, 600 testing). These challenging benchmarks reliably assess the proposed model's robustness and generalization in complex remote sensing scenarios.

B. Parameter Settings

All experiments were conducted on an Ubuntu operating system equipped with an NVIDIA RTX 3080 GPU using PyTorch. PVTv2 served as the backbone, and all input images were resized to 352×352. The network was trained for 60 epochs using a batch size of 8, with optimization performed via the Adam algorithm and an initial learning rate of 1e-4, ensuring stable convergence across datasets.

C. Quantitative Comparison

The proposed AMARA-Net was evaluated against seven state-of-the-art methods (ACCoNet, AESINet [4], ERPNet [5], ADSTNet [6], SFANet [7], TLCKDNet, TSCNet) on the ORSSD and EORSSD datasets. For a clearer assessment of performance progression, comparative models were additionally arranged by publication year.

D. Qualitative Comparison

Visual comparisons against several leading methods were conducted to validate the proposed approach.As illustrated in Figure 2, Our model demonstrates clear advantages in object integrity, boundary sharpness, and background suppression across diverse scenarios, including rivers, roads [8], buildings [9], vehicles, and ships.

TABLE I: COMPARISON OF OUR METHOD WITH 15 OTHER METHODS ON THE ORSSD DATASET

Methods	ORSSD							
	S_α	MAE	E_ξ^{adp}	E_ξ^{mean}	E_ξ^{max}	F_β^{adp}	F_β^{mean}	F_β^{max}
$ACCoNet_{23}$	0.9437	0.0088	0.9721	0.9754	0.9796	0.8806	0.8971	0.9149
$AESINet_{23}$	0.9460	0.0086	0.9707	0.9747	0.9828	0.8666	0.8986	0.9183
$ERPNet_{23}$	0.9254	0.0135	0.9520	0.9566	0.9710	0.8356	0.8745	0.8974
$ADSTNet_{24}$	0.9379	0.0086	0.9785	0.9740	0.9807	0.8979	0.9042	0.9124
$SFANet_{24}$	0.9453	0.0077	0.9765	0.9789	0.9830	0.8984	0.9063	0.9192
$TLCKDNet_{24}$	0.9421	0.0082	0.9696	0.9710	0.9794	0.8719	0.8947	0.9114
$TSCNet_{24}$	0.9427	0.0080	0.9760	0.9805	0.9851	0.8878	0.9030	0.9198
Ours	0.9466	0.0083	0.9803	0.9808	0.9848	0.9003	0.9094	0.9225

The best is highlighted in red, and the second-best is indicated in blue. The number in the lower-right corner of the Methods section corresponds to the publication year.

TABLE II: COMPARISON OF OUR METHOD WITH 15 OTHER METHODS ON THE EORSSD DATASET

Methods	EORSSD							
	S_α	MAE	E_ξ^{adp}	E_ξ^{mean}	E_ξ^{max}	F_β^{adp}	F_β^{mean}	F_β^{max}
$ACCoNet_{23}$	0.9290	0.0074	0.9450	0.9653	0.9727	0.7969	0.8552	0.8837
$AESINet_{23}$	0.9358	0.0079	0.9462	0.9636	0.9751	0.7923	0.8524	0.8838
$ERPNet_{23}$	0.9210	0.0089	0.9228	0.9401	0.9603	0.7554	0.8304	0.8632
$ADSTNet_{24}$	0.9311	0.0065	0.9681	0.9709	0.9769	0.8532	0.8716	0.8804
$SFANet_{24}$	0.9349	0.0058	0.9669	0.9726	0.9769	0.8492	0.8680	0.8833
$TLCKDNet_{24}$	0.9350	0.0056	0.9514	0.9661	0.9788	0.7969	0.8535	0.8843
$TSCNet_{24}$	0.9383	0.0061	0.9688	0.9718	0.9765	0.8509	0.8740	0.8900
Ours	0.9417	0.0051	0.9755	0.9761	0.9806	0.8678	0.8803	0.8939

The best is highlighted in red, and the second-best is indicated in blue. The number in the lower-right corner of the Methods section corresponds to the publication year.

In challenging cases with complex structures (e.g., bridges and curved roads) [10], competing methods often yield incomplete predictions or fragmented edges, whereas our approach consistently preserves structural continuity. For small-scale or scattered targets (e.g., vehicles and ships), most compared models exhibit missed detections or noise artifacts, while ours accurately captures fine-grained details with fewer false responses. Moreover, in scenes dominated by large-scale buildings [11], other methods tend to confuse salient regions with surrounding structures, making precise localization difficult. In contrast, our method maintains clearer separation and preserves the semantic integrity of salient objects.

Overall, the qualitative results confirm that the proposed network not only strengthens global object perception but also improves local detail representation, thereby delivering more reliable and visually coherent predictions than existing methods.

V. DISCUSSION

Aside from the detection results,, computational efficiency is a key consideration for remote sensing applications. To evaluate this aspect, we compared model complexity using the number of parameters and floating-point operations (FLOPs) as metrics, with the results summarized in Table III. The proposed AMARA-Net demonstrates a notably compact design, achieving lower parameter counts and reduced computational demand compared with other state-of-the-art methods. This indicates that the network attains high detection accuracy without incurring redundant computation.Such efficiency facilitates faster inference and lower memory consumption, which are particularly beneficial for large-scale or real-time remote sensing tasks. Overall, AMARA-Net achieves a favorable balance between performance and efficiency, underscoring its

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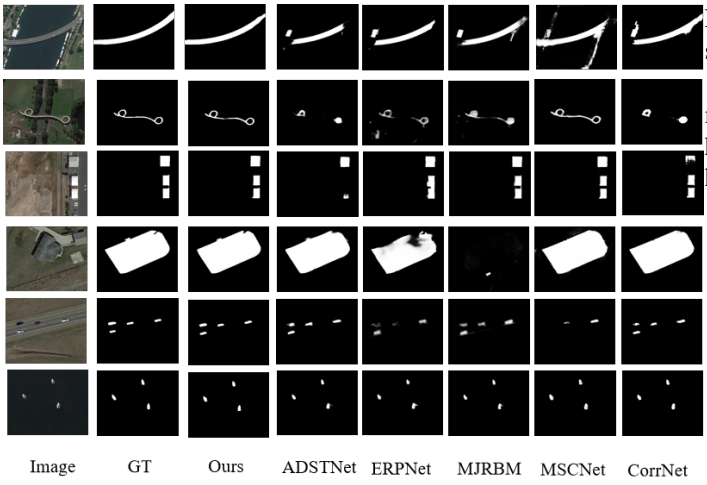


Figure 2: Visual comparison of our method with five state-of-the-art SOD methods

TABLE III: Comparison of model complexity

Methods	Params(M)	FLOps(G)
CorrNet	4.086	21.379
EMFINet	95.086	176
MCCNet	67.652	114
ACCoNet	127	50.422
ERPNet	77.195	171
GeleNet	25.453	6.43
Ours	91.485	19.533

practicality for deployment in resource-constrained environments and its potential for real-world operational applications.

VI. Conclusion

This paper presents AMARA-Net, an Adaptive Multi-scale Aggregation and Residual Attention Network specifically developed to enhance SOD performance in remote sensing scenarios. The proposed framework effectively addresses challenges such as complex backgrounds, weak target contrast, and blurred object boundaries in high-resolution scenes. By combining Residual Channel Attention Blocks (RCABs) with an Adaptive Multi-scale Global Context Aggregation (AMGCA) module, AMARA-Net adaptively captures fine-grained details and global semantics while suppressing redundant noise.

Comprehensive experimental evaluations confirm that AMARA-Net delivers outstanding detection performance compared with existing approaches, sharper boundary delineation, and enhanced robustness. Moreover, the network maintains competitive computational efficiency, achieving a favorable balance between accuracy and resource consumption—an essential consideration for large-scale remote sensing applications. Compared with recent developments,AMARA-Net offers a more lightweight yet effective alternative emphasizing adaptive attention and multi-scale fusion.

In future work, we plan to extend AMARA-Net to multi-modal and multi-temporal remote sensing data, further enhancing its generalization and adaptability to dynamic and heterogeneous observation environments.

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